Project Requirements

STRIDE Threat Model Study — Microservice Data Flows and Structural Threats

1. Software Requirements

The Python helper uses only the Python standard library. Python 3.10 or newer is recommended; no third-party packages are required.

Component	Requirement	Purpose
Python	3.10+ recommended	Runs the threat register generator and score recalculator.
Python libraries	Standard library only	argparse, csv, dataclasses, pathlib, typing.
OWASP Threat Dragon	Recommended	Visual modeling of components, data flows, and trust boundaries.
Spreadsheet tool	Optional	Review and edit the generated CSV threat register.

2. Project Inputs

• Microservice architecture diagram with processes, data stores, external entities, data flows, and trust boundaries.

• Authentication and authorization patterns, protocols, data classifications, and service identities.

• Existing control evidence, threat assumptions, risk criteria, accountable owners, and review cadence.

3. Output

• CSV threat register with STRIDE category, threat scenario, security property, assumed controls, mitigations, likelihood, impact, score, rating, owner, and status.

• Ten illustrative threats across API, identity, payment, audit, database, messaging, secrets, and recovery-related service flows.

• Example score bands: 1–7 Low; 8–14 Moderate; 15–19 High; 20–25 Critical. These are not universal standards.

Security note

This is an educational threat-modeling aid. It does not scan systems or validate actual controls. Treat starter controls as assumptions until verified, and avoid publishing sensitive architecture details in a public repository.